

ANKIT DHITAL

Mechanical Engineer | Design, Fabrication and Controls | Rexburg, ID
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SUMMARY

Mechanical Engineering senior at **BYU-Idaho** (B.S. **Dec 2026**), **CSWP certified**, taking electromechanical designs from concept through **BOM, static FEA**, and functional build. Machines to **0.001 in** on manual mill and lathe, closes control loops on **Arduino**, and documents the work so it repeats. Seeking entry level **product design, mechatronics**, or **manufacturing engineering** roles.

EDUCATION

Brigham Young University-Idaho

Rexburg, ID | Dec 2026

B.S. Mechanical Engineering | **GPA 3.5** | **CSWP Certified** | Coursework: Machine Design, Heat Transfer, Manufacturing Processes II

TECHNICAL SKILLS

CAD and Design: SolidWorks (CSWP), SolidWorks CAM, GD&T, DFM, static FEA, engineering drawings, tolerance stack-ups

Manufacturing: Manual mill and lathe, CNC machining, 3D printing (FDM), laser cutting, plasma cutting, sheet metal forming, welding (basic), dimensional inspection

Controls and Analysis: Arduino, C++, closed-loop control, sensor integration, MATLAB, Python, LabVIEW, Excel, Lean and A3, root cause analysis, SOP and document control

PROJECTS

Thermogate, Closed-Loop Thermal Control System

Mechatronics | Solo build

- Engineered a solo closed-loop temperature system around an **Arduino Uno R3**, integrating **5 subsystems** (mechanical, electrical, firmware, sensing, HMI) in one 3D printed **SolidWorks** housing.
- Held temperature **within 2 °C of setpoint on 90% of 120+ test runs**, driving a servo positioned vent and a DC fan from one control loop against a thermistor reading.
- Diagnosed a **thermistor placement error** reading enclosure heat instead of ambient, and repositioned the sensor outboard.

Briefcase Rat Trap, Electromechanical Rodent Trap

Product Design | Team build

- Validated the disposal door actuator by **static FEA at 3x target load (14.7 N)**, then confirmed **80+ release cycles** without failure.
- Directed CAD, patent research, and **BOM** development; picked the concept by **Pugh matrix** and modeled a **15 part** enclosure built for **\$120 against a \$200 cap, 40% under budget**.
- Engineered a lid activated interlock that cuts live power when the case opens and restores it in **3 seconds**.

4:1 Planetary Gearbox

Machine Design | Fabrication

- Hand calculated a **300% torque increase, 0.61 to 2.44 N-m**, across a **12 part** gearset (12T sun, three 12T planets, 36T ring) over **5 to 6 iterations**.
- Fabricated **PETG** gears and a laser cut acrylic housing into a hand cranked unit, diagnosed a **fastener interference defect** and fixed it with a countersunk chamfer.

Precision Machining and Fabrication, SJSU Machine Shop

Mill | Lathe | Sheet metal

- Machined a **6061 aluminum** hanger to **0.001 in tolerance** on manual mill and lathe, and fabricated an aluminum mailbox to **1/32 in** from flat sheet stock.

EXPERIENCE

Engineering Intern (CI), Basic American Foods

Blackfoot, ID | Apr 2026 to Sep 2026

- Authored and maintained **standard operating procedures** for **Ishida, Iso-Flo**, and **Proctor** dryers, conveyors, and packaging equipment in a high volume food plant.
- Performed **root cause and failure analysis** on production equipment, applying **Lean and A3** problem solving to improve reliability and manufacturability.
- Audited and remapped the plant's **Training Work Instructions (TWIs)** to the equipment and lines they belonged to, working with R&D, **Quality**, and Operations to standardize procedures under **document control**.

CAD Tutor, ME 372 Advanced Design, BYU-Idaho

Rexburg, ID | Jan 2026 to Apr 2026

- Mentored **40+ students** per semester in SolidWorks part and assembly modeling, parametric design, and **DFM**.
- Reviewed **GD&T / ASME Y14.5** drawings and **tolerance stack-ups**, diagnosing mate and feature tree errors before submission.

COMPETITIONS AND ACTIVITIES

ASME Pumpkin Launch: **2nd of 22 teams, 163 yard flight** | **Formula SAE**, San Jose State University